MILITARY STANDARD

MIL-STD-1320-173
(NAVY)

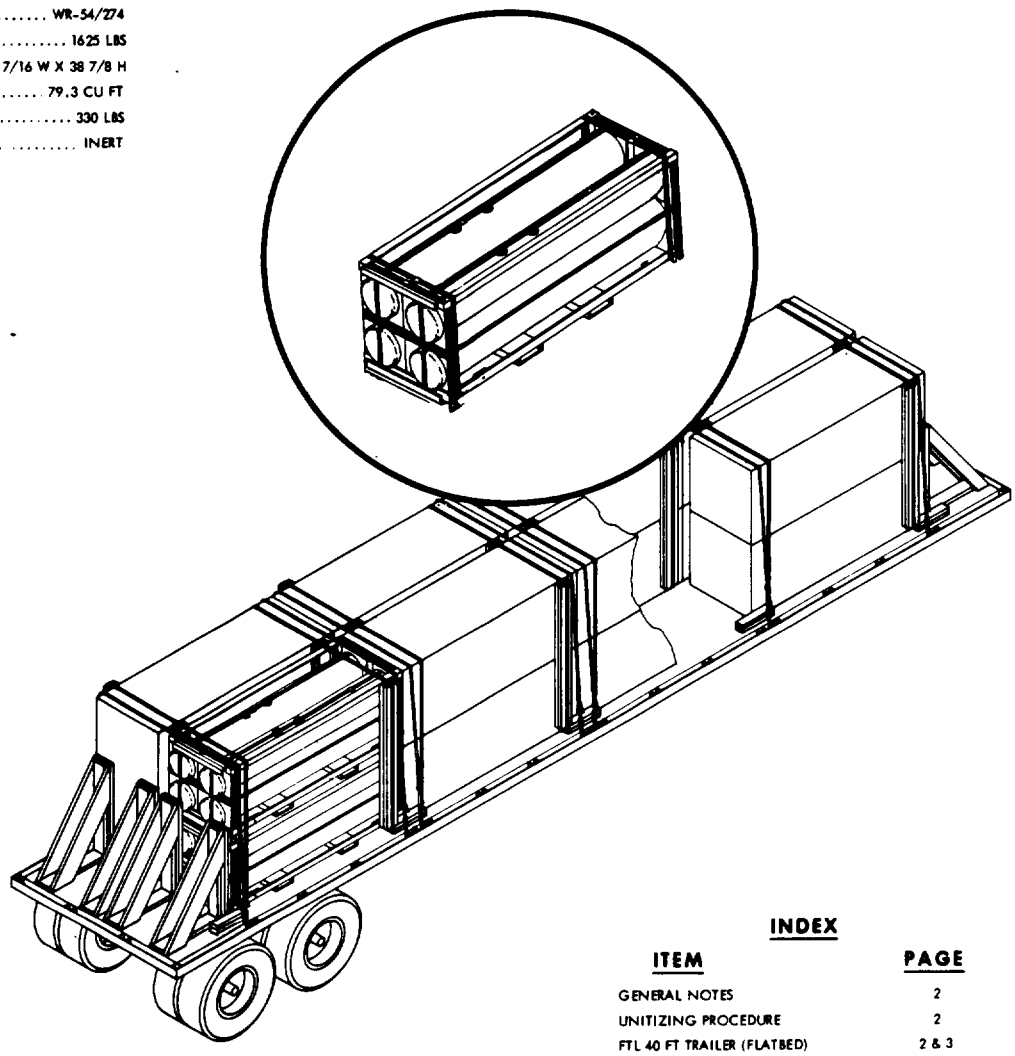
25 MARCH 1976

TRUCKLOADING

DISPENSER, FLARE SUU-25 C/A (EMPTY) ON PALLET, LOADING AND STORAGE MHU-146/E

UNIT LOAD DATA

UNIT LOAD DRAWING	WR-54/274
WEIGHT (APPROX)	1625 LBS
DIMENSIONS	102 L X 35 7/16 W X 38 7/8 H
CUBE	79.3 CU FT
WT OF BOTTOM FRAME	330 LBS
HAZARD CLASSIFICATION	INERT



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FTL 40 FT TRAILER (FLATBED)	2 & 3
LTL 40 FT TRAILER (FLATBED)	4, 5 & 6

NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS, MIL-HDBK-236. (NAVY)

FSC 8140

AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENTS ONLY

M.S. Lucy 3/25/76
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

J.E. Kelly 3/25/76
SIGNATURE AIR SYSCOM, BY DIRECTION DATE

ORIGINATOR Charles McBride 3/25/76
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
WPNSTA EARLE, NEW JERSEY

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GENERAL NOTES

1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING FLEET ISSUE UNIT LOADS OF DISPENSERS, FLARE SUU-25C/A (EMPTY) ON PALLET, LOADING AND STORAGE MHU-146/E. THE DISPENSERS SHALL BE PALLETIZED IN ACCORDANCE WITH WR-54/274.
2. THE PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40 FT FLATBED TRAILERS WITH THE TRAILERS TANDEM AXLES LOCATED IN THE "WESTERN POSITION" (AT THE EXTREME REAR OF THE TRAILER). DO NOT USE TRAILERS WITH METAL FLOORS.
3. ALL TWO HIGH UNIT LOADS SHALL BE UNITIZED AS SHOWN BELOW IN THE UNITIZING PROCEDURE.

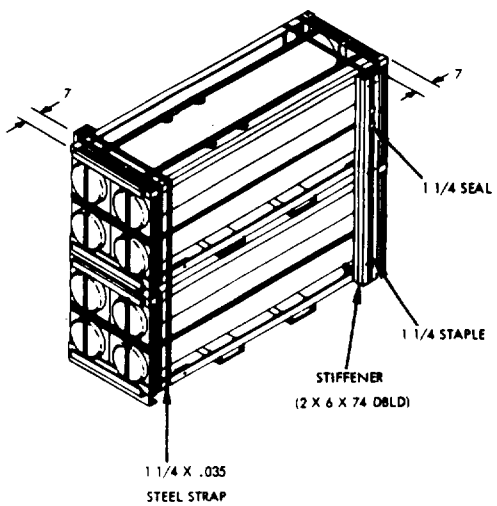
WARNING

THE STACKING AND STRAPPING TOGETHER OF UNIT LOADS IS AUTHORIZED AS PART OF THIS TRUCKLOADING PROCEDURE ONLY. IT IS INTENDED TO ENSURE CONTINUOUS ENGAGEMENT OF THE STACKING FEATURES AND PROVIDE A GOOD TRUCKLOAD. MOVEMENT OF UNIT LOADS IN THIS CONFIGURATION SHALL BE LIMITED TO LOADING AND UNLOADING THE TRAILER.

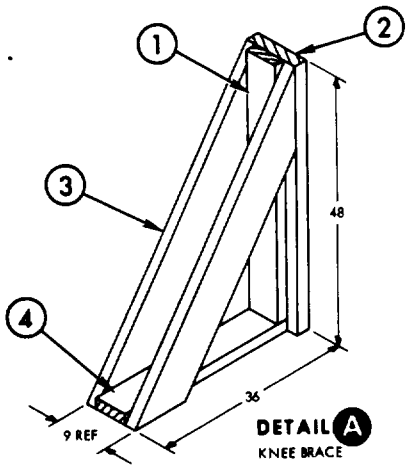
4. KNEE BRACE, DETAIL A (OR D) SHOULD BEAR FIRM AGAINST THE UPPER AND LOWER UNIT LOAD (OR SPACER ASSEMBLY). SHIM IF REQUIRED TO ELIMINATE VOID. SECURE SHIMS BY NAILING SHIM TO KNEE BRACE.
5. CHAINS AND LOADBINDERS SHALL BE USED TO SECURE THE UNIT LOADS TO THE TRAILER AND THEY SHALL MEET THE FOLLOWING CONDITIONS:
   - A. CHAINS AND FITTINGS SHALL BE 3/8 INCH AND LOADBINDERS 3/8 INCH CAPACITY.
   - B. CHAINS, FITTINGS AND LOADBINDERS SHALL HAVE A MINIMUM SAFE WORKING LOAD OF 5000 LBS.
   - C. LOADBINDERS SHALL BE SAFETY WIRED WITH 16 GAUGE SOFT ANNEALED IRON WIRE OR SECURED USING THE END PIECE OF THE 3/8 CHAIN.
6. PRIOR TO LOADING THE TRAILER AND DURING THE PRELOADING INSPECTION REQUIRED BY OP 2165 AND REPORTED ON DD FORM 626, THE CHAINS, FITTINGS AND LOADBINDERS SHALL BE INSPECTED FOR STRETCH, GOUGING, BENT LINKS, WEAR AND ANY OTHER NOTICEABLE DEFECTS. THE INSPECTOR SHALL CONFIRM THAT THE CHAINS AND LOADBINDERS HAVE BEEN INSPECTED AND SHALL SO NOTE IN ITEM NO. 22 OF DD FORM 626. ANY DEFICIENCY SHALL BE CAUSE FOR REJECTION OF A CHAIN OR LOADBINDER.
7. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
8. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO TRAILER IN AN ACCESSIBLE AREA.
9. APPLICABLE MATERIAL SPECIFICATIONS: DUNNAGE LUMBER, MM-L-751; NAILS, FF-N-105, TYPE II. STYLE 10, COMMON BRIGHT; STRAPPING, QQ-S-781, TYPE I CLASS A.
10. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).

UNITIZING PROCEDURE

1. WHEN REQUIRED BY THE FTL OR LTL OF THIS DOCUMENT, THE TWO HIGH STACKED UNIT LOADS SHALL BE STRAPPED TOGETHER AS SHOWN.
2. THE DOUBLED 2 X 6 STIFFENERS SHALL EXTEND FROM THE TOP OF THE UPPER UNIT LOAD TO THE BOTTOM EDGE OF THE BASE FRAME OF THE LOWER UNIT LOAD.
3. THE 1 1/4 STRAPS POSITIONED AS SHOWN ENCIRCLE THE STACKED UNIT LOADS PASSING UNDER THE BOTTOM FRAME OF THE LOWER UNIT LOAD HOLDING THE STIFFENERS IN PLACE.
4. TENSION STRAPS AND SEAL WITH TWO DOUBLE CRIMPED SEALS OR ONE DOUBLE NOTCHED SEAL. SECURE EACH STRAP TO THE STIFFENER WITH THREE 1 1/4 INCH STAPLES.



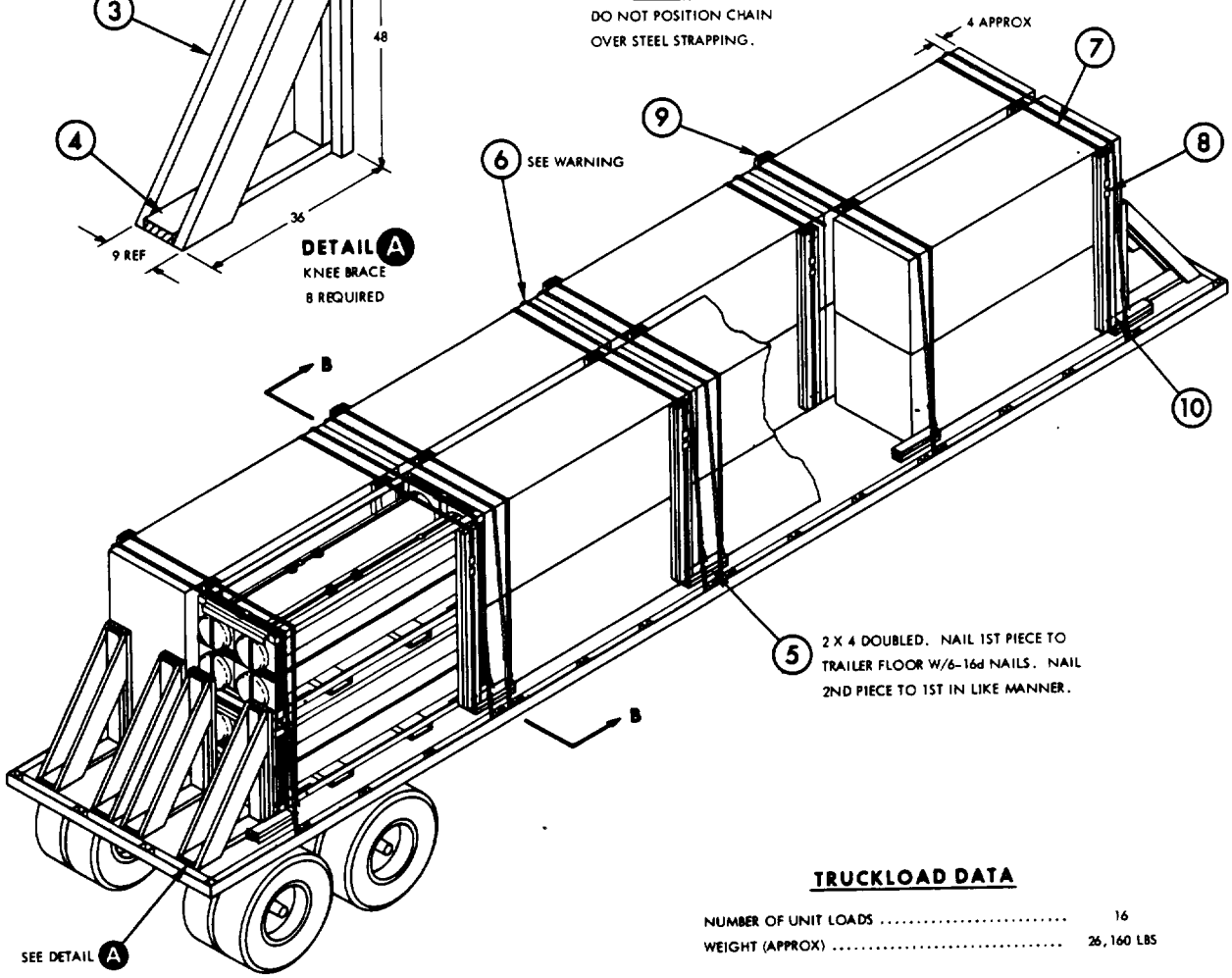
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DETAIL A
KNEE BRACE
8 REQUIRED

POSITION FORWARD UNIT LOADS
36 INCHES FROM FRONT OF TRAILER.

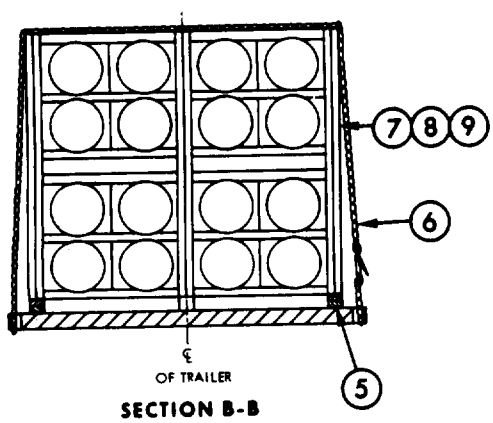
WARNING
DO NOT POSITION CHAIN
OVER STEEL STRAPPING.



5 2 X 4 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W/6-16d NAILS. NAIL 2ND PIECE TO 1ST IN LIKE MANNER.

TRUCKLOAD DATA

NUMBER OF UNIT LOADS	16
WEIGHT (APPROX)	26,160 LBS



SECTION B-B

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
10	STAPLE	FOR 1 1/4" STRAP	48	9	3	1 1/4
9	STIFFENER	2 X 6 X 74	32	-	-	-
8	SEAL	FOR 1 1/4" STRAP	32	-	-	-
7	STEEL STRAP	1 1/4 X .035 X 19 FT 3	16	-	-	-
6	CHAIN & LOADBINDER	3/8	8	-	-	-
5	SLEEPER	2 X 4 X 18	20	SEE FIELD NOTE		
4	BRACE, CLEAT	2 X 6 X 34 1/2	8	TRAILER FLOOR	6	16d
3	BRACE, DIAGONAL	2 X 6 TO SUIT	16	1 & 4	4 PER JOINT	16d
2	BRACE, BACKUP	2 X 10 X 48	8	4	3	16d
1	BRACE, UPRIGHT	2 X 6 X 46 1/2	8	2	6	8d

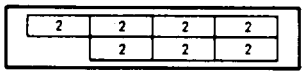
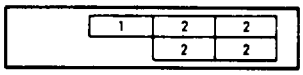
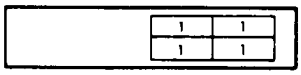
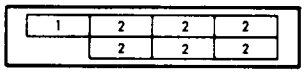
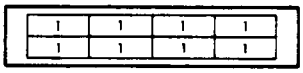
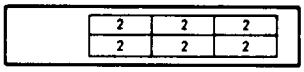
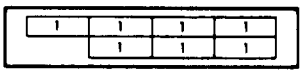
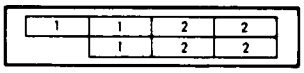
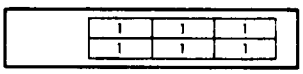
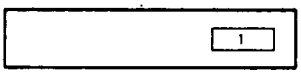
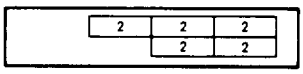
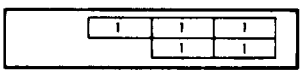
LIST OF MATERIALS & NAILING DATA

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LTL 40 FT TRAILER (FLATBED)

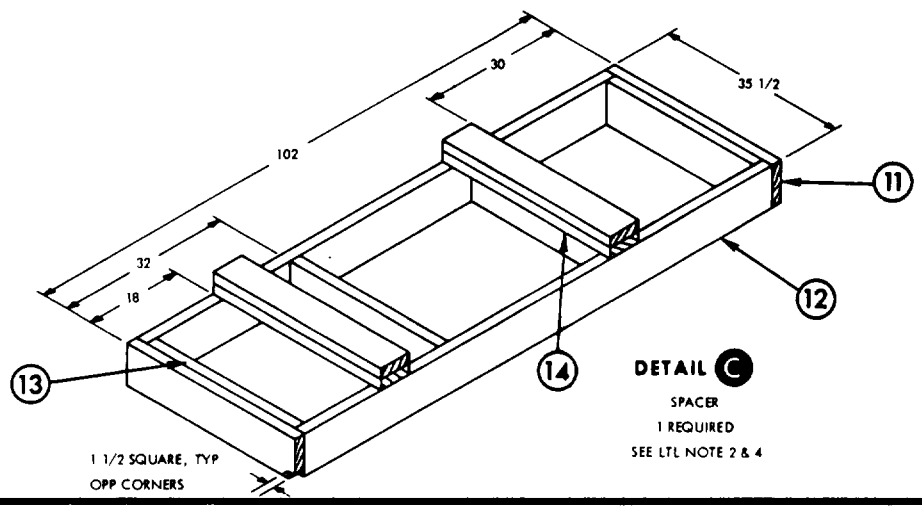
1. WHEN REQUIRED TO SHIP LESS-THAN-TRUCKLOAD QUANTITIES, SELECT THE CORRECT LOAD PATTERN FOR THE NUMBER OF UNIT LOADS TO BE SHIPPED FROM TABLE I AND POSITION THEM ON THE TRAILER ACCORDINGLY. SHIPMENT OF 15 UNIT LOADS IS NOT AUTHORIZED.

TABLE I

NO OF UNIT LOADS	LOAD PATTERN * FWD	NO OF UNIT LOADS	LOAD PATTERN * FWD	NO OF UNIT LOADS	LOAD PATTERN * FWD
14		9		4	
13		8		3	
12		7		2	
11		6		1	
10		5			

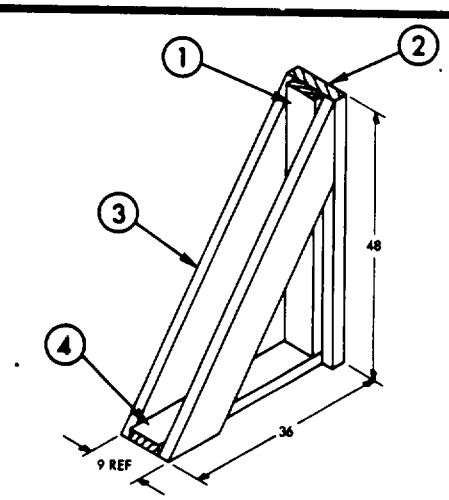
* LOAD PATTERN SHOWS THE FLOOR PLAN FOR THE LOADED TRAILER. THE NUMBER IN THE BLOCKS IS THE NUMBER OF UNIT LOADS IN THE LAYER.

2. WHEN SHIPPING LTL LOADS OF 9, 11 OR 13 UNIT LOADS (SEE TABLE I) WHERE A ONE HIGH UNIT LOAD (S) BACK UP A STACK OF TWO HIGH UNIT LOADS, A SPACER ASSEMBLY MUST BE SECURED TO ALL ONE HIGH LOADS. KNEE BRACES ARE USED TO PREVENT THE LONGITUDINAL MOVEMENT OF THE ONE HIGH UNIT LOADS AND THE SPACER ASSEMBLY. SEE LTL ON PAGE 5.
3. WHEN SHIPPING A LTL OF 8 OR LESS-UNIT LOADS, THE FORE AND AFT BLOCKING SHALL BE ACCOMPLISHED AS SHOWN ON PAGE 6.
4. THE LENGTH OF THE SPACER ASSEMBLY, DETAIL C, SHALL BE SUCH AS TO BEAR AGAINST THE UNIT LOAD IN FRONT OF IT AND THE KNEE BRACE AT THE REAR. SHIM TO FILL VOIDS IF PRESENT USING SUITABLE THICKNESS MATERIAL AND SECURE SHIM IN PLACE BY NAILING TO SPACER ASSEMBLY. SECURE SPACER TO UNIT LOAD TOP ADAPTER RAILS WITH 3/4 X .035 STEEL STRAPS (4 PLACES). SEAL WITH ONE DOUBLE CRIMPED SEAL.
5. THE LTL ON PAGE 5 IS FOR NINE UNIT LOADS AND THE LTL ON PAGE 6 IS FOR FIVE UNIT LOADS. LTL SHIPMENTS ARE NOT LIMITED TO THIS QUANTITY. ACTUAL SHIPPING REQUIREMENTS WILL DETERMINE THE NUMBER OF LOADS TO BE SHIPPED AND THEY SHOULD BE BLOCKED, BRACED AND TIED DOWN IN ACCORDANCE WITH THE APPROPRIATE PARTS OF THIS STANDARD.



DETAIL C
SPACER
1 REQUIRED
SEE LTL NOTE 2 & 4

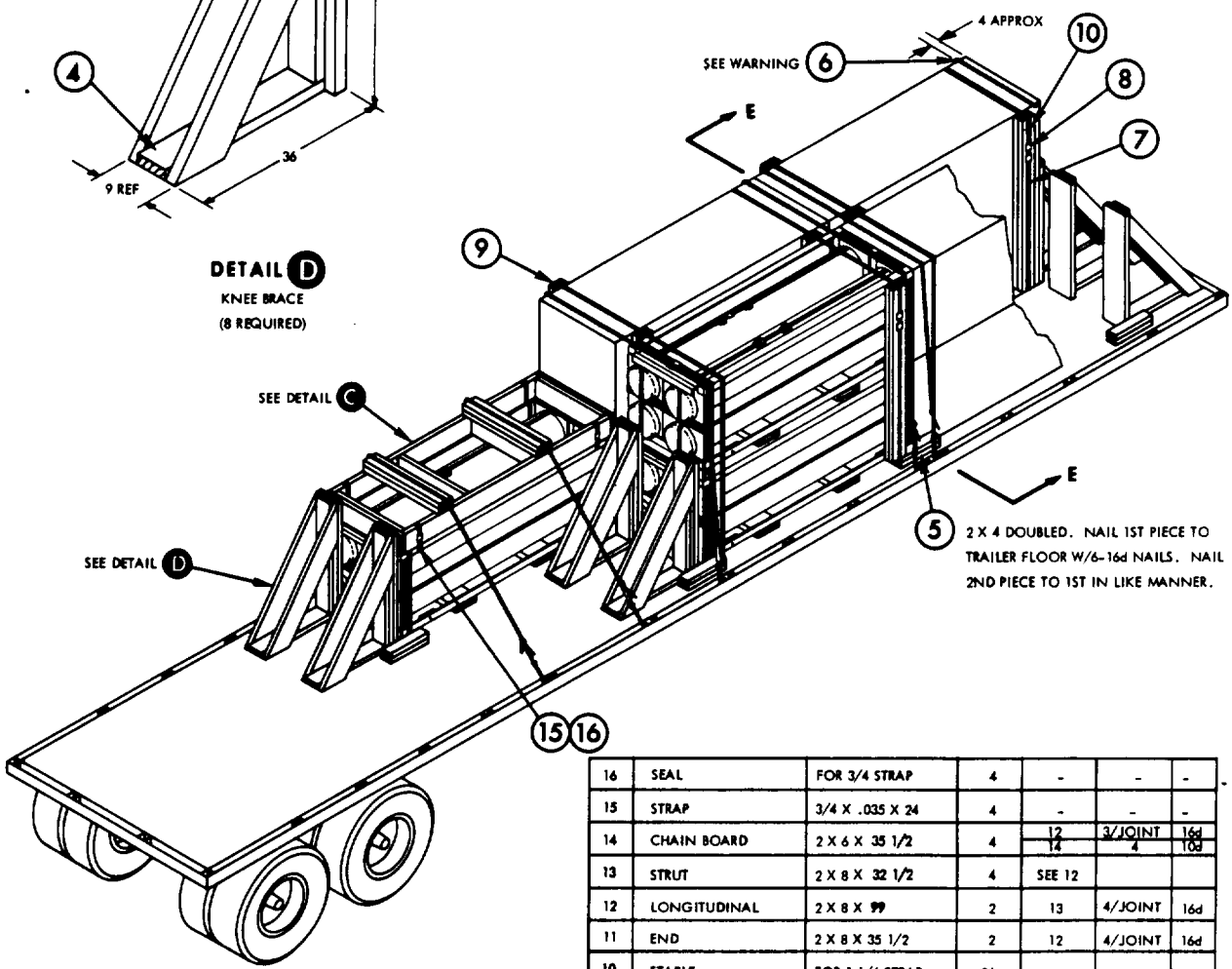
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WARNING

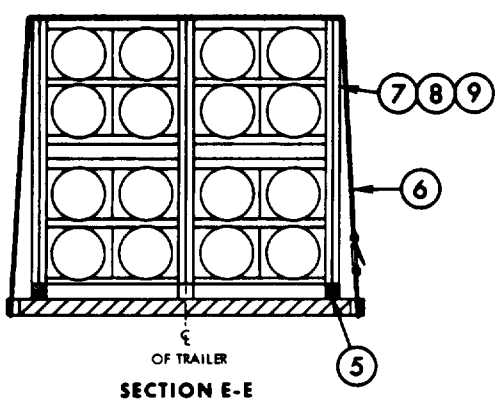
DO NOT POSITION CHAIN OVER STEEL STRAPPING.

POSITION FORWARD UNIT LOADS 36 INCHES FROM FRONT OF TRAILER.



DETAIL D
KNEE BRACE
(8 REQUIRED)

5 2 X 4 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W/6-16d NAILS. NAIL 2ND PIECE TO 1ST IN LIKE MANNER.



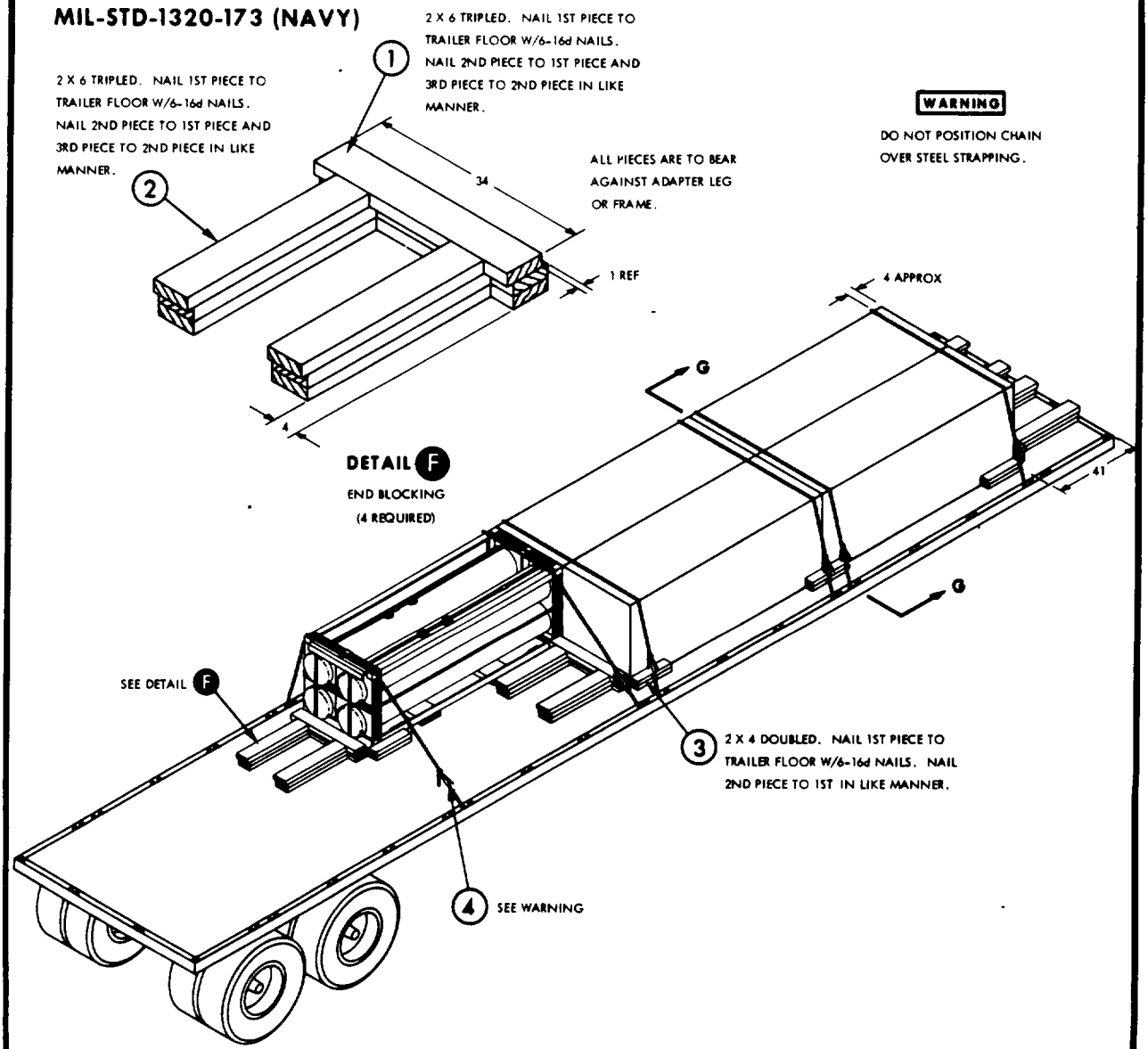
SECTION E-E

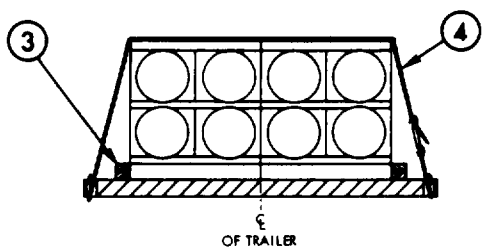
PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER NAILS	SIZE
16	SEAL	FOR 3/4 STRAP	4	-	-	-
15	STRAP	3/4 X .035 X 24	4	-	-	-
14	CHAIN BOARD	2 X 6 X 35 1/2	4	12 14	3/JOINT 4	16d 10d
13	STRUT	2 X 8 X 32 1/2	4	SEE 12		
12	LONGITUDINAL	2 X 8 X 99	2	13	4/JOINT	16d
11	END	2 X 8 X 35 1/2	2	12	4/JOINT	16d
10	STAPLE	FOR 1 1/4 STRAP	24	-	-	-
9	STIFFENER	2 X 6 X 74	16	-	-	-
8	SEAL	FOR 1 1/4 STRAP	16	-	-	-
7	STEEL STRAP	1 1/4 X .035 X 19 FT 3	8	-	-	-
6	CHAIN & LOADBINDER	3/8	6	-	-	-
5	SLEEPER	2 X 4 X 18	16	SEE FIELD NOTE		
4	BRACE, CLEAT	2 X 6 X 34 1/2	8	TRAILER FLOOR	6	16d
3	BRACE, DIAGONAL	2 X 6 TO SUIT	16	1 & 4	4/JOINT	16d
2	BRACE, BACKUP	2 X 10 X 48	8	4	3	16d
1	BRACE, UPRIGHT	2 X 6 X 46 1/2	8	2	6	8d

LIST OF MATERIALS & NAILING DATA

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MIL-STD-1320-173 (NAVY)





SECTION G-G

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
4	CHAIN & LOADBINDER	3/8	6	-	-	-
3	SLEEPER	2 X 4 X 18	16	SEE FIELD NOTE		
2	CLEAT	2 X 6 X 32	24	SEE DETAIL F		
1	CROSSMEMBER	2 X 6 X 34	12	SEE DETAIL F		

LIST OF MATERIALS & NAILING DATA

LTL 40 FT TRAILER (FLATBED)

REVIEW ACTIVITIES:
NAVY-OS, AS

REVIEW ACTIVITIES:
NAVY-OS
(PROJECT NO 8140-N173)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-STD-1320-173 (Navy)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

COMMANDING OFFICER
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NAVAL WEAPONS HANDLING LABORATORY
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